

Adversarial outcomes by protocol and strategy

PBFT	immune $1.0 \times, 0 \text{ VC}$	clean cliff $\phi = 0.40$	fork: 229 unaccountable
Casper FFG	liveness dip 0.60/0.65	graceful decay $\approx 1 - \phi$	$\geq 1/3$ slashable
Snowman	survives $\times 62 / \times 49$	early cliff $\phi = 0.10/0.20$	no fork surface $\varepsilon \approx 5 \times 10^{-15}$
	Delayed voting	Silent non-participation	Equivocation
	robust (liveness held) survives at a latency cost	liveness loss safety failure, accountable	safety break, unaccountable